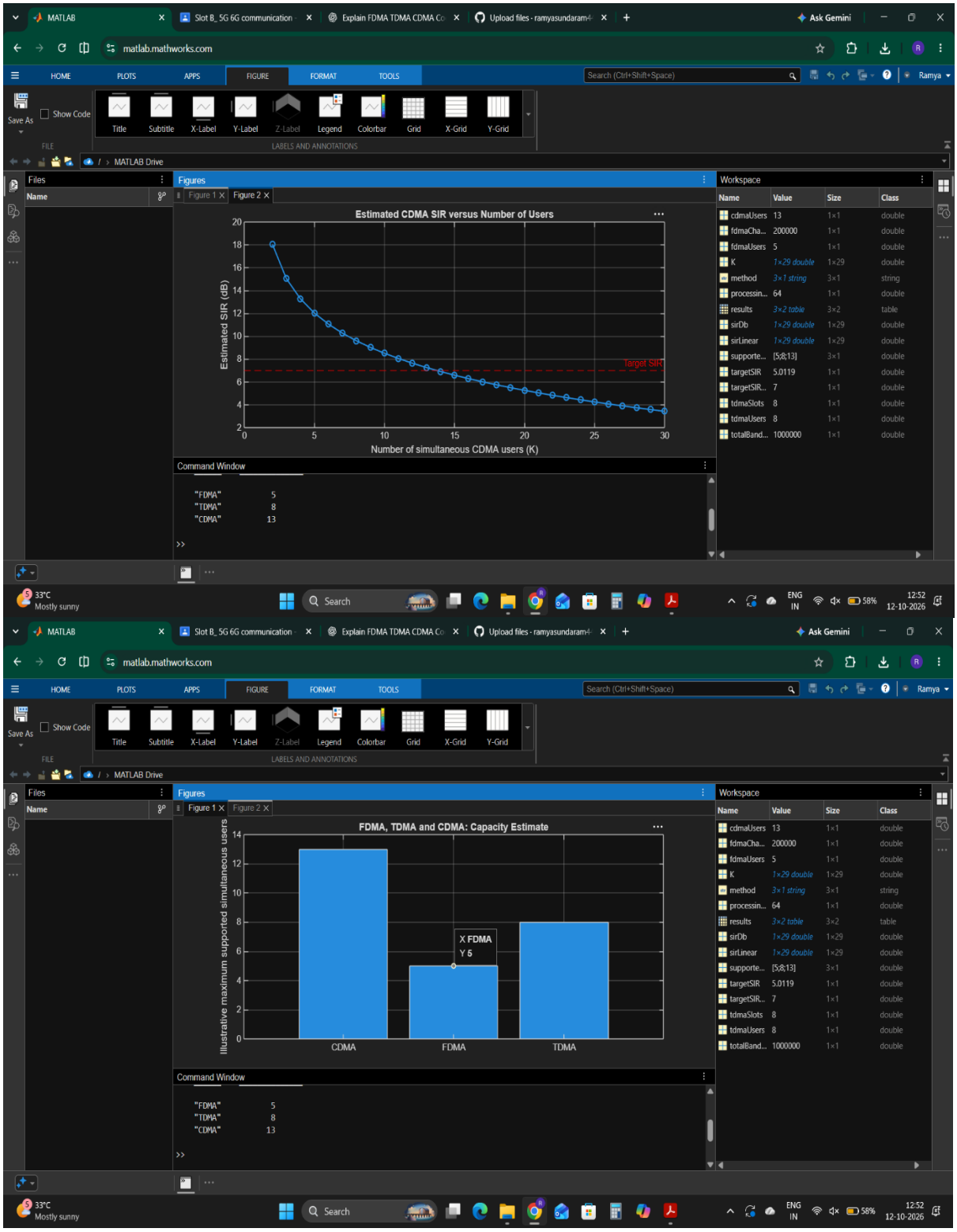
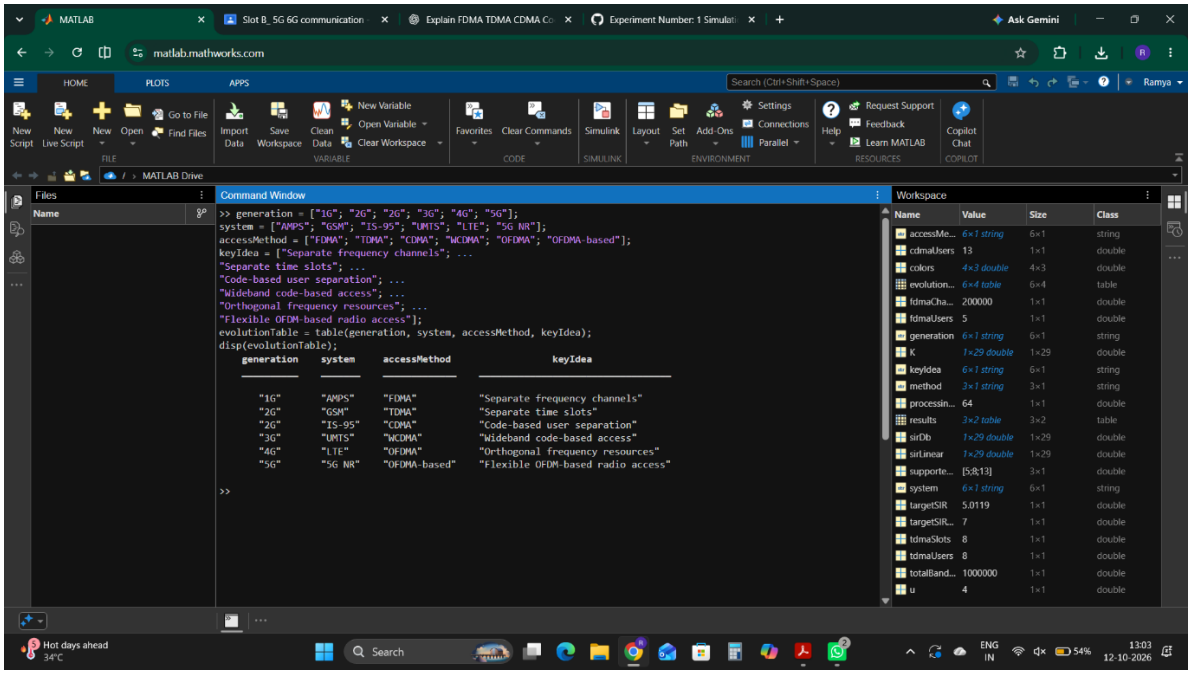
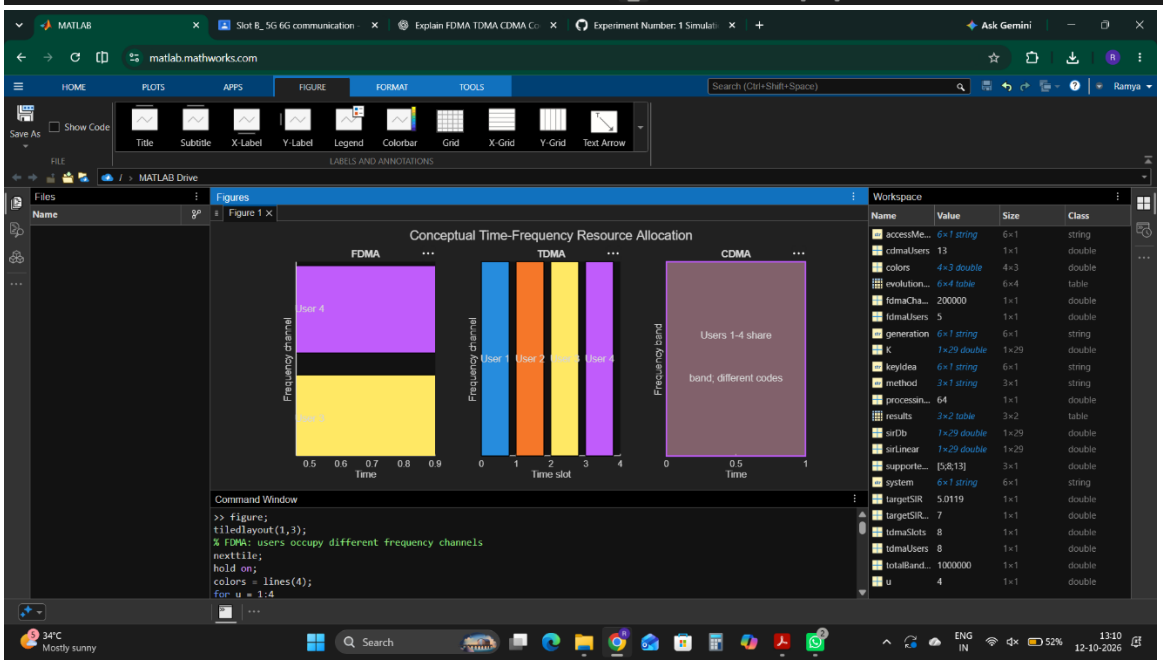
Experiment Number: 1 Simulation and Comparative Analysis of FDMA, TDMA and CDMA: Resource Allocation, User Capacity and Mobile-Generation Context




MATLAB
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HOME
PLOTS
APPS
Command Window
>> generation = ["1G"; "2G"; "2G"; "3G"; "4G"; "5G"];
system = ["AMPS"; "GSM"; "IS-95"; "UMTS"; "LTE"; "5G NR"];
accessMethod = ["FDMA"; "TDMA"; "CDMA"; "WCDMA"; "OFDMA"; "OFDMA-based"];
keyIdea = ["Separate frequency channels"; ...
"Separate time slots"; ...
"Code-based user separation"; ...
"Wideband code-based access"; ...
"Orthogonal frequency resources"; ...
"Flexible OFDM-based radio access"];
evolutionTable = table(generation, system, accessMethod, keyIdea);
disp(evolutionTable);
generation system accessMethod keyIdea
"1G" "AMPS" "FDMA" "Separate frequency channels"
"2G" "GSM" "TDMA" "Separate time slots"
"2G" "IS-95" "CDMA" "Code-based user separation"
"3G" "UMTS" "WCDMA" "Wideband code-based access"
"4G" "LTE" "OFDMA" "Orthogonal frequency resources"
"5G" "5G NR" "OFDMA-based" "Flexible OFDM-based radio access"
>>
Workspace


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Figures
Figure 1
Conceptual Time-Frequency Resource Allocation
FDMA
TDMA
CDMA
Frequency channel
Time
Time slot
Frequency band
Users 1-4 share
band; different codes
Command Window
>> figure;
tiledlayout(1,3);
% FDMA: users occupy different frequency channels
nexttile;
hold on;
colors = lines(4);
for u = 1:4
Workspace